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AI will transform the character of warfare

Technology will make war
faster and more opaque. It
could also prove destabilising



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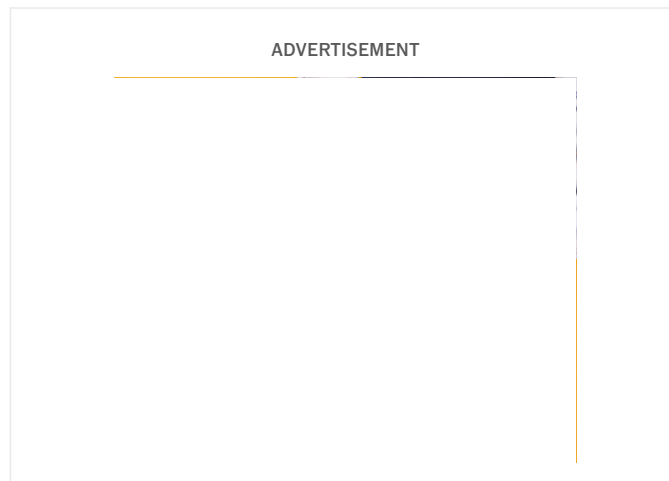
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THE COMPUTER was born in war and by war.
Colossus was built in 1944 to crack Nazi
codes. By the 1950s computers were organising

America's air defences. In the decades that followed, machine intelligence played a small part in warfare. Now it is about to become pivotal. Just as the civilian world is witnessing rapid progress in the power and spread of artificial intelligence (AI), so too must the military world prepare for an onrush of innovation. As much as it transforms the character of war, it could also prove destabilising.

Today's rapid change has several causes. One is the crucible of war itself, most notably in Ukraine. Small, inexpensive chips routinely guide Russian and Ukrainian drones to their targets, scaling up a technology once confined to a superpower's missiles. A second is the recent exponential advance of AI, enabling astonishing feats of object recognition and higher-order problem solving. A third is the rivalry between America and China, in which both see AI as the key to military superiority.



The results are most visible in the advance of intelligent killing machines. Aerial and naval drones have been vital to both sides in Ukraine for spotting and attacking targets. AI's role is as the solution to jamming, because it enables a drone to home in on targets, even if GPS signals or the link to the pilot have been cut. Breaking the connection between pilot and plane should soon let armies deploy far larger numbers of low-cost munitions. Eventually self-directing swarms will be designed to swamp defences.

But what is most visible about military AI is not what is most important. As our briefing explains,

the technology is also revolutionising the command and control that military officers use to orchestrate wars.

On the front line, drones embody just the last and most dramatic link in the kill chain, the series of steps beginning with the search for a target and ending in an attack. AI's deeper significance is what it can do before the drone strikes. Because it sorts through and processes data at superhuman speed, it can pluck every tank out of a thousand satellite images, or interpret light, heat, sound and radio waves to distinguish decoys from the real thing.

Away from the front line, it can solve much larger problems than those faced by a single drone. Today that means simple tasks, such as working out which weapon is best suited to destroying a threat. In due course, "decision-support systems" may be able to grasp the baffling complexity of war rapidly and over a wide area—perhaps an entire battlefield.

The consequences of this are only just becoming clear. AI systems, coupled with autonomous robots on land, sea and air, are likely to find and destroy targets at an unprecedented speed and on a vast scale.

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The speed of such warfare will change the balance between soldier and software. Today, armies keep a man "in the loop", approving each lethal decision. As finding and striking targets is compressed into minutes or seconds, the human may merely "sit on

minutes or seconds, the human may merely “sit on the loop”, as part of a human-machine team. People will oversee the system without intervening in every action.

The paradox is that even as AI gives a clearer sense of the battlefield, war risks becoming more opaque for the people who fight it. There will be less time to stop and think. As the models hand down increasingly oracular judgments, their output will become ever harder to scrutinise without ceding the enemy a lethal advantage. Armies will fear that if they do not give their AI advisers a longer leash, they will be defeated by an adversary who does. Faster combat and fewer pauses will make it harder to negotiate truces or halt escalation. This may favour defenders, who can hunker down while attackers break cover as they advance. Or it may tempt attackers to strike pre-emptively and with massive force, so as to tear down the sensors and networks on which AI-enabled armies will depend.

The scale of AI-based war means that mass and industrial heft are likely to become even more important than they are today. You might think new technology will let armies become leaner. But if software can pick out tens of thousands of targets, armies will need tens of thousands of weapons to strike them. And if the defender has the advantage, attackers will need more weapons to break through.

That is not the only reason AI warfare favours big countries. Drones may get cheaper, but the digital systems that mesh the battlefield together will be fiendishly expensive. Building AI-infused armies will take huge investments in cloud servers able to handle secret data. Armies, navies and air forces that today exist in their own data silos will have to be integrated. Training the models will call for access to vast troves of data.

Which big country does AI favour most? China was once thought to have an advantage, thanks to its pool of data, control over private industry and looser ethical constraints. Yet just now America looks to be ahead in the frontier models that may shape the next generation of military AI. And

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ideology matters: it is unclear whether the armies of authoritarian states, which prize centralised control, will be able to exploit the benefits of a technology that pushes intelligence and insight to the lowest tactical levels.

If, tragically, the first AI-powered war does break out, international law is likely to be pushed to the margins. All the more reason to think today about how to limit the destruction. China should heed America's call to rule out AI control over nuclear weapons, for instance. And once a war begins, human-to-human hotlines will become more important than ever. AI systems told to maximise military advantage will need to be encoded with values and restraints that human commanders take for granted. These include placing an implicit value on human life—how many civilians is it acceptable to kill in pursuing a high-value target?—and avoiding certain destabilising strikes, such as on nuclear early-warning satellites.

The uncertainties are profound. The only sure thing is that AI-driven change is drawing near. The armies that anticipate and master technological advances earliest and most effectively will probably prevail. Everyone else is likely to be a victim. ■

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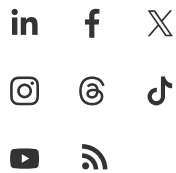
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